

ON A MODIFIED CLASS OF GENERALIZED SKEW NORMAL DISTRIBUTION

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Summary: In this paper we consider a modified class of the generalized skew normal distribution of Arellano-Valle, Gomez and Quintana (2004) and study some of its important properties. A location-scale extension of the distribution is obtained and its parameters are estimated by the method of maximum likelihood. Further, a numerical data application is considered for highlighting the usefulness of the distribution.

1. Introduction

The term 'skew normal' (SN) refers to a parametric class of probability distributions that includes the normal distribution and the properties of this new class resemble those of the normal distribution and appear to justify its name, skew normal. Skew normal distributions have proved to be useful for capturing skewness in data directly without transformation. The skew normal and its extensions might be promising since they preserve the advantage of the normal distribution with the additional benefit of flexibility with regard to skewness and kurtosis. The skew normal distribution and its extensions have been used recently in studies on risk management, capital allocation and goodness of fit. Azzalini (1985) defined the skew normal distribution as follows:

A random variable Z is said to follow a skew normal distribution with parameter $\lambda \in R = (-\infty, \infty)$ if its probability density function (p.d.f.) $g(z; \lambda)$ is of the following form, for $z \in R$.

$$g(z; \lambda) = 2\phi(z)\Phi(\lambda z), \quad (1)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are respectively the p.d.f. and cumulative distribution function (c.d.f.) of a standard normal variate. A distribution with p.d.f. (1) will be denoted by $SND(\lambda)$. $SND(\lambda)$ and its various generalizations are studied by several authors such as Henze (1986), Azzalini (2006), Sharafi

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and Behboodian (2008), Jamalizadeh, Behboodian and Balakrishnan (2008), Venegas, Sanhueza and Gomez (2011), Kumar and Anusree (2011a, 2011b, 2013a, 2013b).

Arellano-Valle et al. (2004) introduced a generalized form of the skew normal distribution, which has the following p.d.f., for $z \in R$

$$g_1(z; \lambda_1, \lambda_2) = 2\phi(z)\Phi\left(\frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}}\right), \quad (2)$$

where $\lambda_1 \in R, \lambda_2 \geq 0$. A distribution with p.d.f. (2) will be denoted by $GSND(\lambda_1, \lambda_2)$. Since the p.d.f. (2) is logconcave, the $GSND(\lambda_1, \lambda_2)$ is not suitable for plurimodal data. Also, the $GSND(\lambda_1, \lambda_2)$ does not accommodate the kurtosis factor of the data properly. So, to mitigate this drawback of the $GSND(\lambda_1, \lambda_2)$, through this paper we propose a modified form of the $GSND(\lambda_1, \lambda_2)$ and termed it as “modified generalized skew normal distribution (MGSND)”. In section 2 we present the definition of the MGSND and discuss some of its important properties. A location-scale extension of the MGSND is considered in section 3 and the maximum likelihood estimation of the parameters of the MGSND is attempted in section 4. Further, in section 5 a real life data application is discussed.

We need the following integral representation in the sequel. For any $a, b \in R$, define

$$\xi(a, b) = \int_a^\infty \int_0^b \frac{1}{2\pi} e^{-\frac{(t^2 + u^2)}{2}} du dt. \quad (3)$$

2. Definition and Properties

We will first present the definition of the MGSND.

Definition 1 A random variable Z is said to have a modified generalized skew normal distribution, (MGSND) if its p.d.f. takes the following form, in which $z \in R, \lambda_1 \in R = (-\infty, \infty), \lambda_2 \geq 0$ and $\alpha \in [0, 1]$,

$$f(z; \lambda_1, \lambda_2, \alpha) = \phi(z) \left[\alpha + 2(1 - \alpha)\Phi\left(\frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}}\right) \right]. \quad (4)$$

Note that $f(z; \lambda_1, \lambda_2, \alpha)$ is a proper p.d.f. since it can be viewed as the mixture of the p.d.f.’s of a standard normal distribution and the $GSND(\lambda_1, \lambda_2)$.

A distribution with p.d.f. (4) will hereafter be denoted by $MGSND(\lambda_1, \lambda_2, \alpha)$. We obtained the probability plots of the $MGSND(\lambda_1, \lambda_2, \alpha)$ for particular values of its parameters in Figures 1 and 2.

Clearly, the $MGSND(\lambda_1, \lambda_2, \alpha)$ has the following special cases, for particular values of its parameters.

1. When $\lambda_1 = 0$ and/or when $\alpha = 1$, the $MGSND(\lambda_1, \lambda_2, \alpha)$ reduces to the standard normal distribution.
2. When $\lambda_1 = \lambda$ and $\lambda_2 = 0$, the $MGSND(\lambda_1, \lambda_2, \alpha)$ reduces to the modified skew normal distribution of Kumar and Anusree (2011b), which we denote hereafter by $MSN(\lambda, \alpha)$.
3. When $\alpha = 0$, $MGSND(\lambda_1, \lambda_2, \alpha)$ reduces to $GSND(\lambda_1, \lambda_2)$ with p.d.f. (2).
4. As λ_1 tends to ∞ and $\alpha = 0$, $MGSND(\lambda_1, \lambda_2, \alpha)$ tends to a half normal distribution.
5. When $\alpha = 0$ and $\lambda_2 = 0$ $MGSND(\lambda_1, \lambda_2, \alpha)$ reduces to $SND(\lambda_1)$.

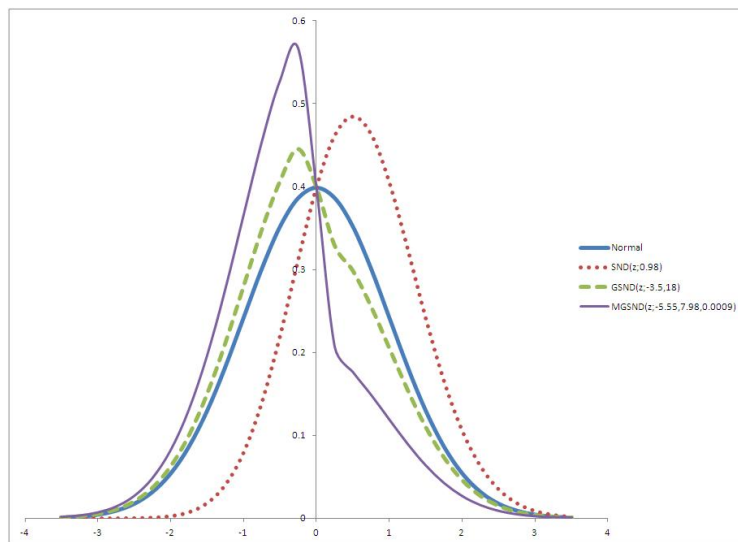


Figure 1: Probability plots of Normal, $SND(0.98)$, $GSND(-3.5,18)$ and $MGSND(-5.55,7.98,0.0009)$.

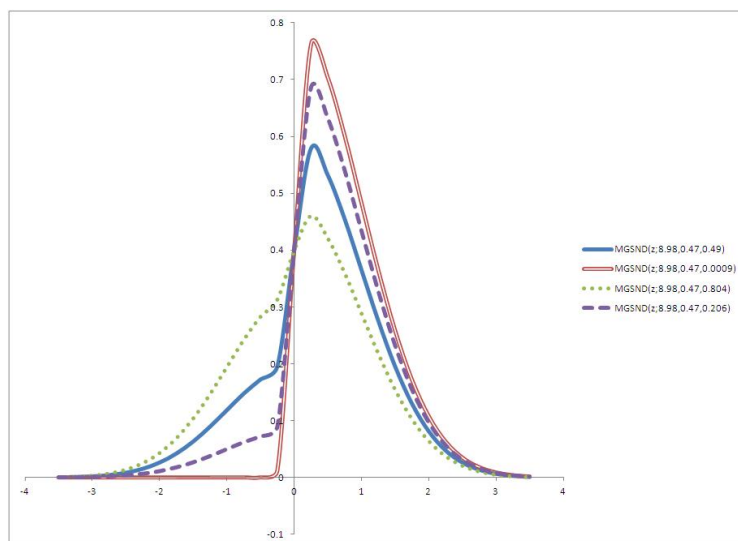


Figure 2: Probability plots of $MGSND(8.98,0.47,\alpha)$ for different values of α .

Result 2.1 If Z follows $MGSND(\lambda_1, \lambda_2, \alpha)$, then $Y_1 = -Z$ follows $MGSND(-\lambda_1, \lambda_2, \alpha)$.

Proof. The p.d.f. $f_1(y_1; \lambda_1, \lambda_2, \alpha)$ of Y_1 is

$$\begin{aligned} f_1(y_1; \lambda_1, \lambda_2, \alpha) &= f(z; \lambda_1, \lambda_2, \alpha) \left| \frac{dz}{dy_1} \right| \\ &= \phi(-y_1) \left[\alpha + 2(1 - \alpha) \Phi \left(\frac{-\lambda_1 y_1}{\sqrt{1 + \lambda_2 y_1^2}} \right) \right] \\ &= f(y_1; -\lambda_1, \lambda_2, \alpha), \end{aligned}$$

which implies that Y_1 follows $MGSND(-\lambda_1, \lambda_2, \alpha)$. ■

Result 2.2 If Z follows $MGSND(\lambda_1, \lambda_2, \alpha)$ then $|Z|$ follows standard half normal distribution.

Proof. Let $Y_2 = |Z|$. Then for any $z \in R$ the p.d.f. $f_2(y_2; \lambda_1, \lambda_2, \alpha)$ of Y_2 is

$$\begin{aligned} f_2(y_2; \lambda_1, \lambda_2, \alpha) &= f(z; \lambda_1, \lambda_2, \alpha) + f(-z; \lambda_1, \lambda_2, \alpha) \\ &= f(z; \lambda_1, \lambda_2, \alpha) + f(z; -\lambda_1, \lambda_2, \alpha), \text{ by Result 2.1} \\ &= \phi(z) \left[\alpha + 2(1 - \alpha) \Phi \left(\frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}} \right) \right] + \\ &\quad \phi(z) \left[\alpha + 2(1 - \alpha) \Phi \left(\frac{-\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}} \right) \right] \\ &= 2\phi(z), \end{aligned}$$

which is the p.d.f. of a half normal random variate since $\phi(\cdot)$ is the p.d.f. of a standard normal variate. ■

Result 2.3 If Z follows $MGSND(\lambda_1, \lambda_2, \alpha)$ then Z^2 follows a chi-square distribution with one degree of freedom.

The proof follows from Result 2.2, since $Z^2 = |Z|^2$, which has a chi-square distribution with one degree of freedom.

Result 2.4 The characteristic function, $\psi(t)$ of a random variable Z following $MGSND(\lambda_1, \lambda_2, \alpha)$ with p.d.f. (4) has the following form, for any $t \in R$ and $i = \sqrt{-1}$.

$$\psi(t) = e^{-\frac{t^2}{2}} \left\{ \alpha + 2(1 - \alpha) E \left(\Phi \left(\frac{\lambda_1 (Y + it)}{\sqrt{1 + \lambda_2 (Y + it)^2}} \right) \right) \right\}, \quad (5)$$

where Y is a standard normal variate.

Proof. Let Z follow $MGSND(\lambda_1, \lambda_2, \alpha)$ with p.d.f. (4). By the definition of characteristic function,

for any $t \in R$ and $i = \sqrt{-1}$,

$$\begin{aligned}\psi(t) &= E(e^{itZ}) \\ &= \left[\alpha \int_{-\infty}^{\infty} e^{itz} \phi(z) dz + 2(1-\alpha) \int_{-\infty}^{\infty} e^{itz} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \Phi\left(\frac{\lambda_1 z}{\sqrt{1+\lambda_2 z^2}}\right) dz \right] \\ &= e^{-\frac{t^2}{2}} \left[\alpha + 2(1-\alpha) \int_{-\infty}^{\infty} \phi(z-it) \Phi\left(\frac{\lambda_1 z}{\sqrt{1+\lambda_2 z^2}}\right) dz \right].\end{aligned}$$

On substituting $z-it = y$, $\psi(t)$ reduces to

$$\psi(t) = e^{-\frac{t^2}{2}} \left[\alpha + 2(1-\alpha) \int_{-\infty}^{\infty} \phi(y) \Phi\left(\frac{\lambda_1(y+it)}{\sqrt{1+\lambda_2(y+it)^2}}\right) dy \right],$$

which gives (5). ■

Result 2.5 The cumulative distribution function (c.d.f.) of a random variable Z following $MGSND(\lambda_1, \lambda_2, \alpha)$ is given by

$$F(z; \lambda_1, \lambda_2, \alpha) = \Phi(z) - 2(1-\alpha) \xi\left(z; \frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right), \quad (6)$$

where $\xi\left(z; \frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right)$ is as defined in (3).

Proof. Let Z be a random variable with p.d.f. (4) and $F(\cdot)$ be the c.d.f. of Z . Then by the definition of a c.d.f. we have

$$\begin{aligned}F(z; \lambda_1, \lambda_2, \alpha) &= \int_{-\infty}^z f(t; \lambda_1, \lambda_2, \alpha) dt \\ &= \int_{-\infty}^z \left[\alpha + 2(1-\alpha) \Phi\left(\frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right) \right] \phi(t) dt \\ &= \alpha \Phi(z) + 2(1-\alpha) \int_{-\infty}^z \left[\frac{1}{2} + \int_0^{\frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}} \phi(u) du \right] \phi(t) dt\end{aligned}$$

which implies (6). ■

As a consequence of Result 2.2, we can see that the even moments of the $MGSND(\lambda_1, \lambda_2, \alpha)$ and standard normal distribution are identical. In order to obtain an expression for odd order raw moments of the $MGSND(\lambda_1, \lambda_2, \alpha)$, we establish the following result.

Result 2.6 If Z follows $MGSND(\lambda_1, \lambda_2, \alpha)$, then for $k = 0, 1, 2, \dots$

$$E(Z^{2k+1}) = \left[\alpha \pi^{-1} 2^{k+\frac{1}{2}} \Gamma(k+1) + 2(1-\alpha) \Lambda_k(\lambda_1, \lambda_2) \right], \quad (7)$$

in which

$$\Lambda_k(\lambda_1, \lambda_2) = \int_0^\infty \frac{u^k}{\sqrt{2\pi}} e^{-\frac{u}{2}} \Phi\left(\frac{\lambda_1 \sqrt{u}}{\sqrt{1+\lambda_2 u}}\right) du, \quad (8)$$

for $\lambda_1 \in R$ and $\lambda_2 \geq 0$.

Proof. By definition of raw moments,

$$E(Z^{2k+1}) = \int_{-\infty}^\infty z^{2k+1} f(z) \left[\alpha + 2(1-\alpha) \Phi\left(\frac{\lambda_1 z}{\sqrt{1+\lambda_2 z^2}}\right) \right] dz. \quad (9)$$

On substituting $z^2 = u$ in (9), we get,

$$\begin{aligned} E(Z^{2k+1}) &= \int_0^\infty u^{k+\frac{1}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u}{2}} \left[\alpha + 2(1-\alpha) \Phi\left(\frac{\lambda_1 \sqrt{u}}{\sqrt{1+\lambda_2 u}}\right) \right] \frac{1}{2\sqrt{u}} du \\ &= \left[\alpha \int_0^\infty u^k \frac{e^{-\frac{u}{2}}}{\sqrt{2\pi}} du + 2(1-\alpha) \int_0^\infty u^k \frac{e^{-\frac{u}{2}}}{\sqrt{2\pi}} \Phi\left(\frac{\lambda_1 \sqrt{u}}{\sqrt{1+\lambda_2 u}}\right) du \right] \end{aligned}$$

which leads to (7). ■

Result 2.7 The reliability function $R(t; \lambda_1, \lambda_2, \alpha)$ of the $MGSND(\lambda_1, \lambda_2, \alpha)$ is given by

$$R(t; \lambda_1, \lambda_2, \alpha) = 1 - \Phi(t) + 2(1-\alpha) \xi\left(t; \frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right),$$

where $\xi\left(t; \frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right)$ is as defined in (3).

The proof follows from the definition of reliability function and Result 2.5.

Result 2.8 The failure rate $r(t; \lambda_1, \lambda_2, \alpha)$ of the $MGSND(\lambda_1, \lambda_2, \alpha)$ is given by

$$r(t; \lambda_1, \lambda_2, \alpha) = \frac{\phi(t) \left[\alpha + 2(1-\alpha) \Phi\left(\frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right) \right]}{\left[[1 - \phi(t)] + 2(1-\alpha) \xi\left(t; \frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right) \right]}.$$

The proof follows from the definition of failure rate and Result 2.5.

Result 2.9 The mean residual life function $M(t; \lambda_1, \lambda_2, \alpha)$ of $MGSND(\lambda_1, \lambda_2, \alpha)$ is given by:

$$\begin{aligned} M(t; \lambda_1, \lambda_2, \alpha) &= R(t; \lambda_1, \lambda_2, \alpha)^{-1} \left[e^{-\frac{t^2}{2}} + 2(1-\alpha) e^{-\frac{t^2}{2}} \Phi\left(\frac{\lambda_1 t}{\sqrt{1+\lambda_2 t^2}}\right) + \right. \\ &\quad \left. 2(1-\alpha) \sqrt{2\pi} \xi^*(t; \lambda_1, \lambda_2) \right] - t, \end{aligned} \quad (10)$$

in which $\xi^*(t; \lambda_1, \lambda_2) = \int_t^\infty \phi(z) \left[\frac{d}{dz} \left(\int_0^{\frac{\lambda_1 z}{\sqrt{1+\lambda_2 z^2}}} \phi(u) du \right) \right] dz$.

Proof. By definition, the mean residual life function (MRLF), of Z is

$$\begin{aligned} M(t; \lambda_1, \lambda_2, \alpha) &= E(Z - t | Z > t) \\ &= E(Z | Z > t) - t \end{aligned}$$

where

$$\begin{aligned} E(Z | Z > t) &= R(t; \lambda_1, \lambda_2, \alpha)^{-1} \left[\int_t^\infty \alpha z \phi(z) dz + \int_t^\infty 2(1 - \alpha) z \phi(z) \Phi\left(\frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}}\right) dz \right] \\ &= R(t; \lambda_1, \lambda_2, \alpha)^{-1} \sqrt{2\pi}^{-1} \left[\alpha e^{-\frac{t^2}{2}} + 2(1 - \alpha) e^{-\frac{t^2}{2}} \Phi\left(\frac{\lambda_1 t}{\sqrt{1 + \lambda_2 t^2}}\right) \right. \\ &\quad \left. + 2(1 - \alpha) \sqrt{2\pi} \int_t^\infty \phi(z) \left[\frac{d}{dz} \left(\int_0^{\frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}}} \phi(u) du \right) \right] dz \right], \end{aligned}$$

which implies (10). ■

Result 2.10 The p.d.f. of $MGSND(\lambda_1, \lambda_2, \alpha)$ is log concave at least for the following cases, in which

$$\kappa(\lambda_1, \lambda_2) = \frac{2(1 - \alpha) \lambda_1^2 \phi'(d)}{[\alpha + 2(1 - \alpha) \Phi(d)][1 + \lambda_2 z^2]^3},$$

$$\eta(\lambda_1, \lambda_2) = \frac{[2(1 - \alpha)]^2 \lambda_1^2 \phi(d)^2}{[\alpha + 2(1 - \alpha) \Phi(d)]^2 [1 + \lambda_2 z^2]^3}$$

and

$$\zeta(\lambda_1, \lambda_2) = \frac{2(1 - \alpha) \lambda_1 \lambda_2 z \phi(d)}{[\alpha + 2(1 - \alpha) \Phi(d)][1 + \lambda_2 z^2]^{5/2}}$$

with

$$d = \frac{\lambda_1 z}{\sqrt{1 + \lambda_2 z^2}}.$$

1. case 1: For all $\lambda_1 \leq 0$ and $\lambda_2 z < 0$.
2. case 2: For all $\lambda_1 \leq 0$ and $\lambda_2 z > 0$ if $3\zeta(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2) + \kappa(\lambda_1, \lambda_2)$.
3. case 3: For all $\lambda_1 \geq 0$ and $\lambda_2 z > 0$, if $\kappa(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2)$.
4. case 4: For all $\lambda_1 \geq 0$ and $\lambda_2 z < 0$ if $\kappa(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2)$.

Proof. Taking the logarithm on both sides of the p.d.f. of $MGSND(\lambda_1, \lambda_2, \alpha)$ given in (4) and differentiating twice with respect to z , we get the following,

$$\frac{d^2}{dz^2} [\log(f(z; \lambda_1, \lambda_2, \alpha))] = -1 - \eta(\lambda_1, \lambda_2) + \kappa(\lambda_1, \lambda_2) - 3\zeta(\lambda_1, \lambda_2). \quad (11)$$

Clearly $\eta(\lambda_1, \lambda_2)$ is positive for all $\alpha \in [0, 1]$, $\lambda_1 \in (-\infty, \infty)$ and $\lambda_2 > 0$. $\kappa(\lambda_1, \lambda_2)$ is non-negative when $\lambda_1 \geq 0$ and negative for $\lambda_1 < 0$. $\zeta(\lambda_1, \lambda_2)$ is non-negative when either $\lambda_1 \geq 0$ and $\lambda_2 z > 0$ or $\lambda_1 \leq 0$ and $\lambda_2 z < 0$ and negative when either $\lambda_1 < 0$ or $\lambda_2 z < 0$. Now by using (11) we can conclude that the p.d.f given in (4) is logconcave in the cases listed in 1, 2, 3 and 4. ■

As a consequence of Result 2.10 we have the following results.

Result 2.11 The p.d.f. of $MGSND(\lambda_1, \lambda_2, \alpha)$ is unimodal at least for the following cases

1. case 1: For all $\lambda_1 \leq 0$ and $\lambda_2 z < 0$.
2. case 2: For all $\lambda_1 \leq 0$ and $\lambda_2 z > 0$ if $3\zeta(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2) + \kappa(\lambda_1, \lambda_2)$.
3. case 3: For all $\lambda_1 \geq 0$ and $\lambda_2 z > 0$, if $\kappa(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2)$.
4. case 4: For all $\lambda_1 \geq 0$ and $\lambda_2 z < 0$ if $\kappa(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2) < 1 + \eta(\lambda_1, \lambda_2)$.

Result 2.12 The p.d.f. of $MGSND(\lambda_1, \lambda_2, \alpha)$ is plurimodal at least for the following cases

1. case 1: For all $\lambda_1 \leq 0$ and $\lambda_2 z > 0$ if $3\zeta(\lambda_1, \lambda_2) \geq 1 + \eta(\lambda_1, \lambda_2) + \kappa(\lambda_1, \lambda_2)$.
2. case 2: For all $\lambda_1 \geq 0$ and $\lambda_2 z > 0$, if $\kappa(\lambda_1, \lambda_2) \geq 1 + \eta(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2)$.
3. case 3: For all $\lambda_1 \geq 0$ and $\lambda_2 z < 0$ if $\kappa(\lambda_1, \lambda_2) + 3\zeta(\lambda_1, \lambda_2) \geq 1 + \eta(\lambda_1, \lambda_2)$.

3. Location-Scale Extension

In this section, we define the location-scale extension of the $MGSND(\lambda_1, \lambda_2, \alpha)$ and derive some of its important properties.

Definition 2 If Z follows $MGSND(\lambda_1, \lambda_2, \alpha)$, then the distribution of the random variable $X = \mu + \sigma Z$, for any $\mu \in R = (-\infty, \infty)$ and $\sigma > 0$, is called the location-scale extension of the $MGSND(\lambda_1, \lambda_2, \alpha)$. For $\lambda_1 \in R$, $\lambda_2 \geq 0$, $\alpha \in [0, 1]$, the p.d.f. of $MGSND(\lambda_1, \lambda_2, \alpha)$ is

$$f(x; \mu, \sigma, \lambda_1, \lambda_2, \alpha) = \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right) \left\{ \alpha + 2(1 - \alpha) \Phi \left[\frac{\lambda_1 (x - \mu)}{\sqrt{\sigma^2 + \lambda_2 (x - \mu)^2}} \right] \right\}, \quad (12)$$

in which μ and σ are the location and scale parameters respectively. For convenience of notation, we denote this family by $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$.

Result 3.1 The characteristic function, $\psi_X(t)$ of a random variable Z following $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$ with p.d.f. (12) has the following form, for any $t \in R$ and $i = \sqrt{-1}$ is,

$$\psi_X(t) = e^{\mu it - \frac{t^2 \sigma^2}{2}} \left\{ \alpha + 2(1 - \alpha) E_Y \left[\Phi \left(\frac{\lambda_1 (Y + \sigma it)}{\sqrt{1 + \lambda_2 (Y + \sigma it)^2}} \right) \right] \right\}, \quad (13)$$

where Y is an $N(\mu, \sigma^2)$ variate.

Result 3.2 If Y follows $N(\lambda_1, \lambda_2)$ and the conditional distribution of X given $Y = y$ follows $MSN(y, \alpha)$ then X follows $MGSND(\lambda_1, \lambda_2^2, \alpha)$.

Proof. Let $h_{11}(x, y)$ denote the joint p.d.f. of X and Y , $h_{10}(x|y)$ denote the conditional p.d.f. of X given $Y = y$ and $h_2(y)$ denote the marginal p.d.f. of Y . Then the marginal p.d.f. $h_1(\cdot)$ of X is the following, for any $x \in R$, $\alpha \in [0, 1]$ and $\lambda_1, \lambda_2 \in R$,

$$\begin{aligned} h_1(x; \lambda_1, \lambda_2, \alpha) &= \int_y h_{11}(x, y) dy \\ &= \int_y h_{10}(x|y) h_2(y) dy \\ &= \int_y \frac{1}{\lambda_2} \phi(x) [\alpha + 2(1 - \alpha) \Phi(xy)] \phi\left(\frac{y - \lambda_1}{\lambda_2}\right) dy. \end{aligned} \quad (14)$$

On substituting $\frac{y - \lambda_1}{\lambda_2} = z$ in (14) we get

$$\begin{aligned} h_1(x; \lambda_1, \lambda_2, \alpha) &= \phi(x) \int_{-\infty}^{\infty} [\alpha + 2(1 - \alpha) \Phi(\lambda_2 x z + \lambda_1 x)] \phi(z) dz \\ &= \phi(x) E[\alpha + 2(1 - \alpha) \Phi(\lambda_2 x Z + \lambda_1 x)], \end{aligned}$$

since $Z = \frac{Y - \lambda_1}{\lambda_2}$ follows $N(0, 1)$ with p.d.f. $\phi(z)$. Thus we have the following

$$h_1(x; \lambda_1, \lambda_2, \alpha) = \phi(x) \left[\alpha + 2(1 - \alpha) \Phi\left(\frac{\lambda_1 x}{\sqrt{1 + \lambda_2^2 x^2}}\right) \right],$$

which shows that X follows $MGSND(\lambda_1, \lambda_2^2, \alpha)$. ■

Result 3.3 If Y follows $N(0, \tau^2)$ and the conditional distribution of X given $Y = y$ follows $MGSND^{LS}(\mu, \sigma; y, y^2, \alpha)$, then X follows $N(\mu, \sigma^2)$ and the conditional distribution of Y given $X = x$ follows $MGSND^{LS}(0, \tau; \tau(x - \mu)\sigma^{-1}, \alpha)$.

Proof. Let $h_{10}(x|y)$, $h_{01}(y|x)$, $h_1(x)$ and $h_2(y)$ respectively denote the conditional p.d.f. of X given $Y = y$, conditional p.d.f. of Y given $X = x$, marginal p.d.f. of X and the marginal p.d.f. of Y . Then the marginal p.d.f. $h_1(x)$ of X can be written as,

$$\begin{aligned} h_1(x) &= \int_y h_{10}(x|y) h_2(y) dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right) \left[\alpha + 2(1 - \alpha) \Phi\left(\frac{y(x - \mu)}{\sqrt{\sigma^2 + y^2(x - \mu)^2}}\right) \right] \frac{1}{\tau} \phi\left(\frac{y}{\tau}\right) dy. \end{aligned} \quad (15)$$

On substituting $t = \frac{y}{\tau}$ in (15) we get,

$$\begin{aligned} h_1(x) &= \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right) \left[\alpha + 2(1 - \alpha) \int_{-\infty}^{\infty} \phi(t) \Phi\left(\frac{t \tau (x - \mu)}{\sqrt{\sigma^2 + t^2 \tau^2 (x - \mu)^2}}\right) dt \right] \\ &= \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right) \end{aligned}$$

which implies X follows $N(\mu, \sigma^2)$. Now the conditional distribution of Y given $X = x$ can be written as

$$\begin{aligned}
 h_{01}(y|x) &= \frac{h_{10}(x|y)}{h_1(x)} \\
 &= \frac{\frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) \left[\alpha + 2(1-\alpha) \Phi\left(\frac{y(x-\mu)}{\sqrt{\sigma^2 + y^2(x-\mu)^2}}\right) \right] \frac{1}{\tau} \phi\left(\frac{y}{\tau}\right)}{\frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right)} \\
 &= \frac{1}{\tau} \phi\left(\frac{y}{\tau}\right) \left[\alpha + 2(1-\alpha) \Phi\left(\frac{y(x-\mu)}{\sqrt{\sigma^2 + y^2(x-\mu)^2}}\right) \right] \\
 &= \frac{1}{\tau} \phi\left(\frac{y}{\tau}\right) \left[\alpha + 2(1-\alpha) \Phi\left(\frac{y\tau(x-\mu)}{\sigma\sqrt{\tau^2 + \frac{y^2\tau^2}{\sigma^2}(x-\mu)^2}}\right) \right].
 \end{aligned}$$

Thus the conditional distribution of Y given $X = x$ follows $MGSND^{LS}(0, \tau; \tau(x-\mu)\sigma^{-1}, \alpha)$. ■

Result 3.4 Let the conditional distribution of X given $Y_1 = y_1$ and $Y_2 = y_2$ follows $MGSND(\mu, \sigma; y_1, y_2, \alpha)$ in which $\mu \in R$, $\sigma > 0$, $y_1 \in R$, $y_2 \in R$, and $\alpha \in [0, 1]$. If the conditional distribution of Y_1 given $Y_2 = y_2$ follows $N(0, \sigma_1^2)$ and Y_2 has p.d.f. f_{Y_2} , then X and Y_2 are independent and X follows $N(\mu, \sigma^2)$. Further the conditional distribution of Y_1 given $X = x$, $Y_2 = y_2$ follows the extended modified skew normal distribution denoted by $EMSN(0, \sigma_1^2; \varpi, \alpha)$ of Kumar and Anusree (2011b), where $\varpi = \frac{\sigma_1(x-\mu)}{\sqrt{\sigma^2 + y_2(x-\mu)^2}}$.

Proof. The conditional distribution of X given Y_2 is

$$\begin{aligned}
 f_{X|Y_2}(x) &= \int_{-\infty}^{\infty} \frac{f_{X,Y_1,Y_2}(x, y_1, y_2)}{f_{Y_2}(y_2)} dy_1 \\
 &= \int_{-\infty}^{\infty} f_{X|Y_1,Y_2}(x) f_{Y_1|Y_2}(y_1) dy_1 \\
 &= \int_{-\infty}^{\infty} \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) \left\{ \alpha + 2(1-\alpha) \Phi\left[\frac{y_1(x-\mu)}{\sqrt{\sigma^2 + y_2(x-\mu)^2}}\right] \right\} \frac{1}{\sigma_1} \phi\left(\frac{y_1}{\sigma_1}\right) dy_1.
 \end{aligned} \tag{16}$$

On substituting $\frac{y_1}{\sigma_1} = t$ in (16) we get,

$$\begin{aligned}
 f_{X|Y_2}(x) &= \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) \left[\alpha + 2(1-\alpha) \int_{-\infty}^{\infty} \Phi\left(\frac{\sigma_1(x-\mu)t}{\sqrt{\sigma^2 + y_2(x-\mu)^2}}\right) \phi(t) dt \right] \\
 &= \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right),
 \end{aligned}$$

which is independent of Y_2 .

Hence X and Y_2 are independent and X follows $N(\mu, \sigma^2)$. Then for $y_1 \in R$,

$$\begin{aligned} f_{Y_1|X, Y_2}(y_1) &= \frac{f(x, y_1, y_2)}{f(x)f(y_2)} \\ &= \frac{f(x|y_1, y_2)f(y_1|y_2)f(y_2)}{f(x)f(y_2)} \\ &= \frac{\frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) \left\{ \alpha + 2(1-\alpha) \Phi \left[\frac{y_1(x-\mu)}{\sqrt{\sigma^2 + y_2(x-\mu)^2}} \right] \right\} \frac{1}{\sigma_1} \phi\left(\frac{y_1}{\sigma_1}\right)}{\frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right)} \\ &= \frac{1}{\sigma_1} \phi\left(\frac{y_1}{\sigma_1}\right) \left\{ \alpha + 2(1-\alpha) \Phi \left[\frac{y_1 \sigma_1 (x-\mu)}{\sigma_1 \sqrt{\sigma^2 + y_2 (x-\mu)^2}} \right] \right\}, \end{aligned}$$

which shows that Y_1 given $X = x, Y_2 = y_2$ follows $EMSN(0, \sigma_1^2; \sigma, \alpha,)$. ■

Note that the Results 3.2 to 3.4 are relevant in the context of Bayesian inference since in each of these results normal priors yield normal marginals.

4. Maximum Likelihood Estimation

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population following $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$ with p.d.f. (12). Now the likelihood of the sample is

$$L = \left(\frac{1}{\sqrt{2\pi}} \right)^n \frac{e^{-\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2}}}{\sigma^n} \prod_{i=1}^n [\alpha + 2(1-\alpha) \Phi(C_i)],$$

where $C_i = \left(\frac{\lambda_1 (x_i - \mu)}{\sqrt{\sigma^2 + \lambda_2 (x_i - \mu)^2}} \right)$.

Taking logarithm on both sides we have

$$\begin{aligned} \log L &= n \log \left(\frac{1}{\sqrt{2\pi}} \right) - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} - n \log \sigma \\ &\quad + \sum_{i=1}^n \log [\alpha + 2(1-\alpha) \Phi(C_i)]. \end{aligned} \tag{17}$$

On differentiating (17) with respect to $\mu, \sigma^2, \lambda_1, \lambda_2$ and α and equating to zero we get the normal equations (18), (19), (20), (21) and (22) as given below,

$\frac{\partial \log L}{\partial \mu} = 0$ implies

$$\sum_{i=1}^n \frac{(x_i - \mu)}{\sigma^2} = \sum_{i=1}^n \frac{2(1-\alpha) \phi(C_i)}{[\alpha + 2(1-\alpha) \Phi(C_i)]} \frac{\lambda_1 \sigma}{[\sigma^2 + \lambda_2 (x_i - \mu)^2]^{\frac{3}{2}}}, \tag{18}$$

$\frac{\partial \log L}{\partial \sigma^2} = 0$ implies

$$\frac{n}{2\sigma^2} = \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^4} + \sum_{i=1}^n \frac{2(1-\alpha)\phi(C_i)}{\alpha + 2(1-\alpha)\Phi(C_i)} \frac{-1}{2} \left[\sigma^2 + \lambda_2 (x_i - \mu)^2 \right]^{-\frac{3}{2}} \times \lambda_1 (x_i - \mu), \quad (19)$$

$\frac{\partial \log L}{\partial \lambda_1} = 0$ implies

$$\sum_{i=1}^n \frac{2(1-\alpha)\phi(C_i)}{\alpha + 2(1-\alpha)\Phi(C_i)} \left(\frac{(x_i - \mu)}{\sqrt{\sigma^2 + \lambda_2 (x_i - \mu)^2}} \right) = 0, \quad (20)$$

$\frac{\partial \log L}{\partial \lambda_2} = 0$ implies

$$\sum_{i=1}^n \frac{2(1-\alpha)\phi(C_i)}{\alpha + 2(1-\alpha)\Phi(C_i)} \lambda_1 (x_i - \mu) \left(\frac{(x_i - \mu)^2}{2(\sigma^2 + \lambda_2 (x_i - \mu)^2)^{\frac{3}{2}}} \right) = 0, \quad (21)$$

and $\frac{\partial \log L}{\partial \alpha} = 0$ implies

$$\sum_{i=1}^n \frac{1 - 2\Phi(C_i)}{\alpha + 2(1-\alpha)\Phi(C_i)} = 0. \quad (22)$$

If we put

$$\frac{\phi(C_i)}{[\alpha + 2(1-\alpha)\Phi(C_i)]} \frac{1}{\left(\sigma^2 + \lambda_2 (x_i - \mu)^2 \right)^{\frac{3}{2}}} = W(x_i)$$

and

$$\frac{\Phi(C_i)}{\alpha + 2(1-\alpha)\Phi(C_i)} = V(x_i)$$

in the above normal equations, we get the following system of equations:

$$\sum_{i=1}^n \frac{(x_i - \mu)}{\sigma^2} = \lambda_1 \sigma \sum_{i=1}^n 2(1-\alpha)W(x_i) \quad (23)$$

$$\frac{n}{\sigma^2} = \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^4} + 2(1-\alpha)\lambda_1 \sum_{i=1}^n W(x_i) (x_i - \mu) \quad (24)$$

$$\alpha \sum_{i=1}^n W(x_i) (x_i - \mu) [\sigma^2 + \lambda_2 (x_i - \mu)^2] = 0 \quad (25)$$

$$\sum_{i=1}^n 2(1-\alpha)\lambda_1 (x_i - \mu)^3 W(x_i) = 0 \quad (26)$$

$$\sum_{i=1}^n \frac{1}{\alpha + 2(1-\alpha)\Phi(C_i)} - 2 \sum_{i=1}^n V(x_i) = 0. \quad (27)$$

With the help of MATHCAD software, we have verified that the second order partial derivatives of the log-likelihood function with respect to the parameters $\mu, \sigma, \lambda_1, \lambda_2$ and α gives negative values for all values of $\mu \in R, \sigma > 0, \lambda_1 \in R, \lambda_2 \geq 0$ and $\alpha \in [0, 1]$. Therefore on solving the non-linear system of equations from (23) to (27) by using mathematical softwares such as MATHCAD, MATHEMATICA, MATLAB etc. one can obtain the maximum likelihood estimators of the parameters of the $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$.

5. A Data Application and Concluding Remarks

For carrying out a numerical illustration here we consider the data matrices related to milk production of 28 cows taken from Bhuyan (2005) and the variable under study is the daily milk production in kilogram(the variable recorded for three times milking cows). The data is as given below:

34.6 27.7 29.2 25.3 27.6 37.9 32.6 32 30.7 29.6 38.3 32.9 30.8 32.2 32.9 28.1 33.9 28.6 28.1 35.9 34.8 40.3 30.9 34.4 19.8 25.8 37.3 32.4.

The numerical results obtained are presented in Table 1, which includes the estimated values of the parameters with values of the corresponding Kolmogorov Smirnov Statistic (KSS) , the Akaike's Information Criterion (AIC) and the Bayesian Information Criterion (BIC) values for different models such as normal, $SND(\mu, \sigma; \lambda)$, $GSND(\mu, \sigma; \lambda_1, \lambda_2)$ and $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2)$. A graph of the fitted models and the histogram of the data are presented in Figure 3. It is clear from Table 1 and

Table 1: Estimated values of the parameters with corresponding values of KSS , AIC and BIC for the models normal $N(\mu, \sigma)$, $SND(\mu, \sigma; \lambda)$, $GSND(\mu, \sigma; \lambda_1, \lambda_2)$ and $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$.

Distribution	$N(\mu, \sigma)$	$SND(\mu, \sigma; \lambda)$	$GSND(\mu, \sigma; \lambda_1, \lambda_2)$	$MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$
$\hat{\mu}$	31.59286	32.95	31.593	31.593
$\hat{\sigma}$	4.464421	4.587	4.384	4.384
$\hat{\lambda}$	-	-0.394	-	-
$\hat{\lambda}_1$	-	-	2.487×10^{-5}	-8.302×10^{-2}
$\hat{\lambda}_2$	-	-	0.562	0.018
$\hat{\alpha}$	-	-	-	0.5135
KSS	0.063	0.0625	0.06	0.05
AIC	166.23	168.18	170.21	138.91
BIC	168.9	172.17	175.54	145.6

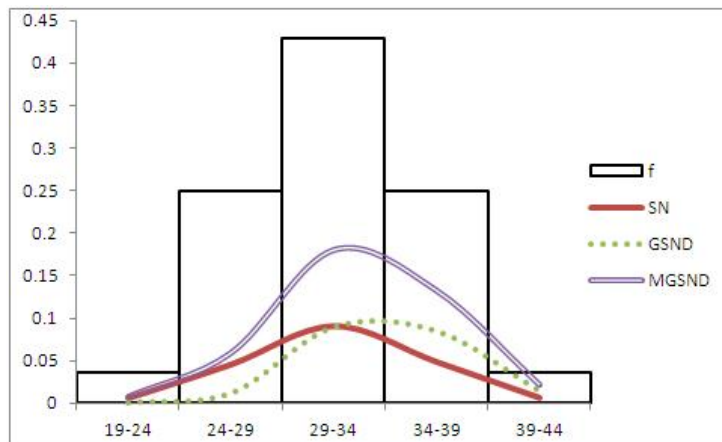


Figure 3: Histogram of the data and the fitted distributions.

Figure 3 that the $MGSND^{LS}(\mu, \sigma; \lambda_1, \lambda_2, \alpha)$ is a more appropriate model to the data compared to the existing models - normal, $SND(\mu, \sigma; \lambda)$ and $GSND(\mu, \sigma; \lambda_1, \lambda_2)$.

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